

Inside LC0's Mind: A Visual Guide

An Illustrated Step-by-Step Exposition of Neural MCTS

Chess Speak Out Loud Project

August 8, 2026

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Part 0: How to read these pictures

Every search-tree diagram in this document is drawn using a fixed visual grammar. The meaning of every border, color, arrow, and bar is defined here once and remains constant throughout the guide.

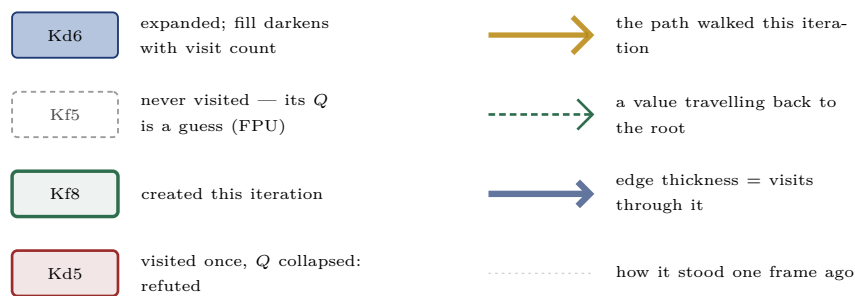


Figure 1: FIG-0.1: The visual grammar legend. Node borders, fill saturations, path highlights, and edge thicknesses maintain byte-identical definitions across all tree frames.

Key idea

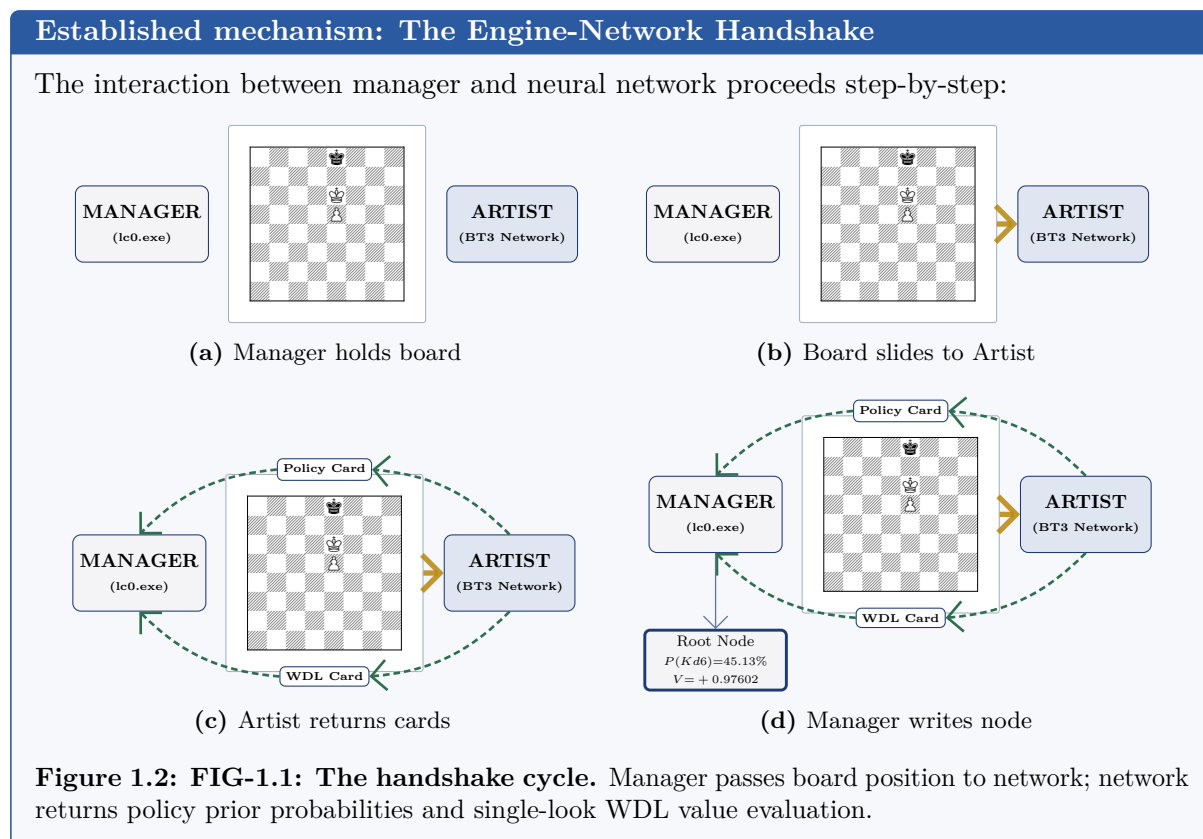
Solid blue borders represent expanded nodes evaluated by the network; dashed grey borders represent unvisited nodes whose Q value is an FPU (First Play Urgency) guess. The thick green border marks a node created on the current iteration, while red fill indicates a refuted move where Q collapsed.

Chapter 1

The Two Partners

The search engine operates as a partnership between two distinct components: the **Manager** (`lc0.exe`), which maintains tree bookkeeping and decision selection, and the **Artist** (the BT3 neural network), which evaluates positions in single forward passes.

1.1 The Handshake



1.2 Anatomy of a Tree Node

Established mechanism: Search Tree Node Anatomy

Each node in the MCTS search tree stores both measured network feedback and running statistics:

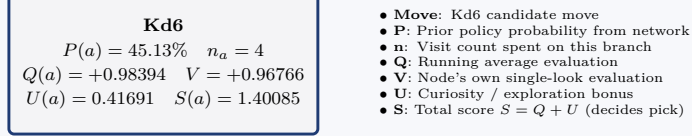


Figure 1.3: FIG-1.2: Node Anatomy. Prior probability $P(a)$, visit count n_a , running average evaluation $Q(a)$, single-pass value V , curiosity bonus $U(a)$, and selection score $S(a) = Q(a) + U(a)$.

1.3 The Initial Network Evaluation

Real engine output: Initial Position Assessment

For our subject endgame position (4k3/8/4K3/4P3/8/8/8/8 w - - 0 1), one single forward pass produces initial policy priors and win-draw-loss assessment:

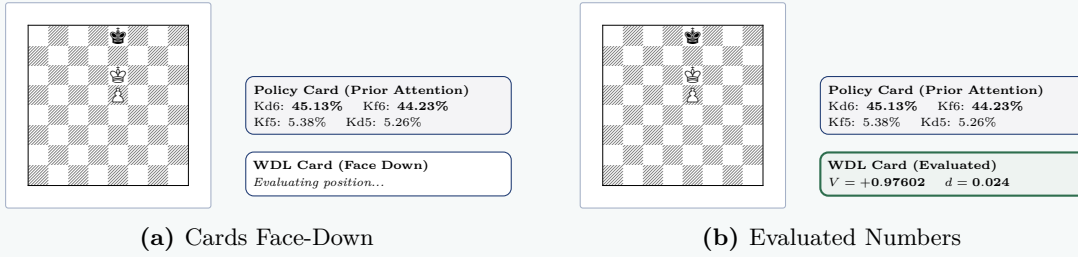


Figure 1.5: FIG-1.3: Policy and WDL cards for K+P endgame. The policy places 89.36% of attention on winning moves (Kd6 45.13%, Kf6 44.23%), but still leaves 10.64% on drawing moves (Kf5 5.38%, Kd5 5.26%). Initial root value $V = +0.97602$, $d = 0.024$.

Key idea

The network policy is an informed initial instinct, not an unassailable oracle. 89.36% of its attention went to winning moves, but 10.64% remained on moves that throw away the win. The search exists to verify and refine this instinct.

Chapter 2

The Growing Tree

The spine of the guide is an eight-iteration search on our four-legal-move endgame position. Because there are only four legal moves at the root, every single node in the tree can be drawn explicitly at fixed coordinates without schematic shortcuts.

2.1 The Eight Search Iterations

We trace the search iteration by iteration. A FIG-2.x frame captures two moments: the **tree** displays the state *after* iteration x completes (new leaf evaluated, visit counts incremented, backpropagated values), while the **S-bar strip** below displays the scores compared *before* iteration x 's pick — the numbers that caused the selection. The tallest bar ($S = Q + U$) carries gold highlight and matches the candidate move marked with the gold selection ring in the tree above.

Real engine output: Iteration 0

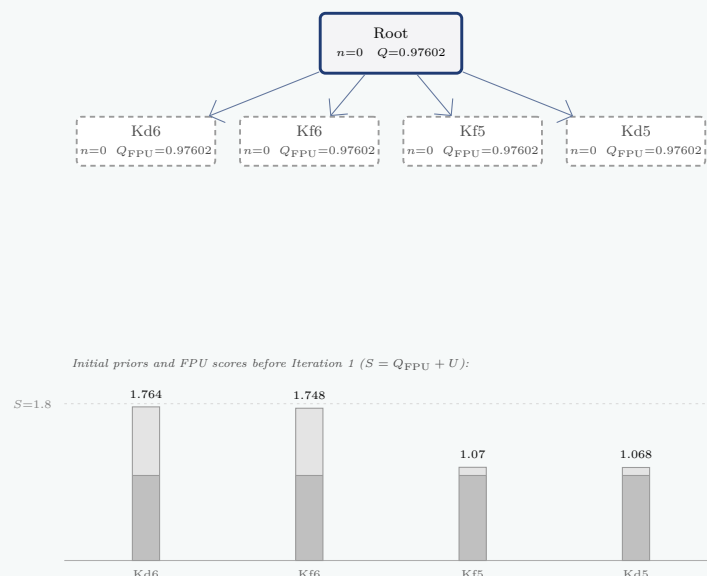


Figure 2.1: FIG-2.0: Iteration 0 (Starting State). All four children unvisited (dashed grey). All Q values are equal to $Q_{FPU} = +0.97602$. Ranking is purely decided by policy priors.

Real engine output: Iteration 1

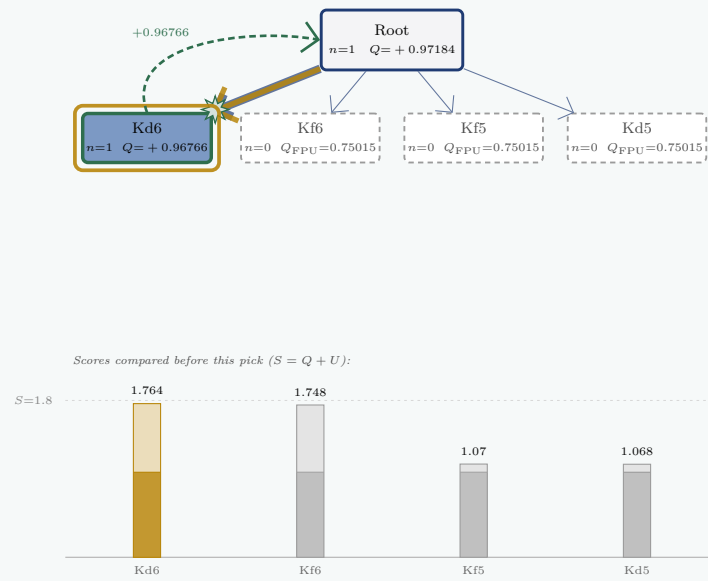


Figure 2.2: FIG-2.1: Iteration 1. Kd6 selected ($S = 1.76358$). Expanded leaf returns $+0.96766$. Kd6 becomes solid green burst; backpropagation updates root to $Q = +0.97184$.

Real engine output: Iteration 2

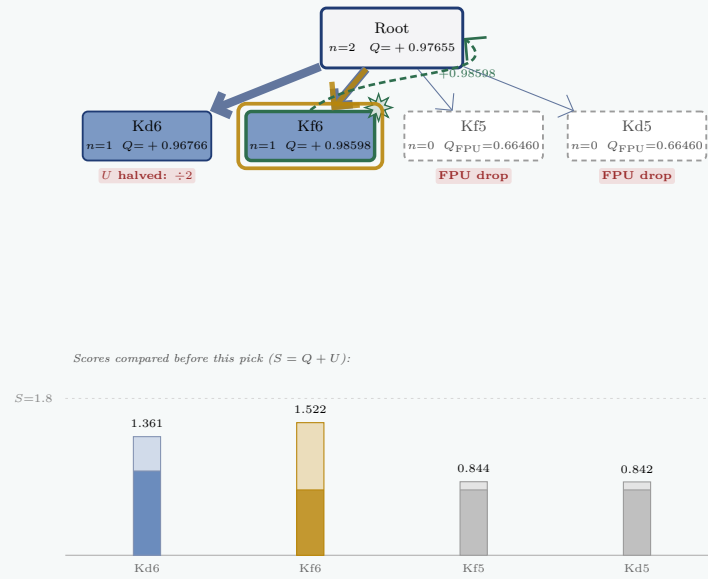


Figure 2.3: FIG-2.2: Iteration 2 (The PUCT Demonstration). Kd6 has the best measured score (+0.96766 vs pre-pick FPU 0.75015, grey bars) yet is **not** selected. Its U halved ($1 + 0 \rightarrow 1 + 1$ denominator), allowing Kf6 ($S = 1.52205$) to take the lead. Post-pick FPU updates to 0.66460 (dashed nodes).

Real engine output: Iteration 3

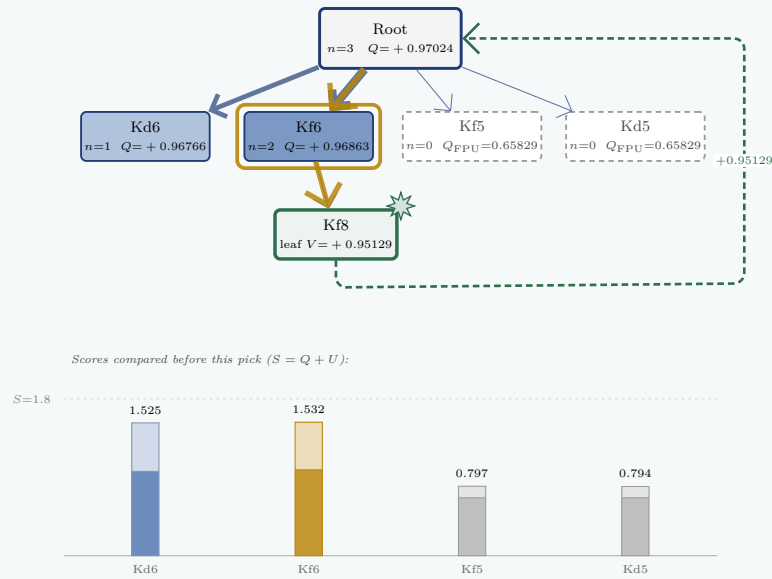


Figure 2.4: FIG-2.3: Iteration 3 (Selection Recurses). Kf6 selected ($S = 1.53183$). Selection does not stop at root; it recurses to Kf6's child node Kf8 (depth 2). Leaf value +0.95129 backs up.

Real engine output: Iteration 4

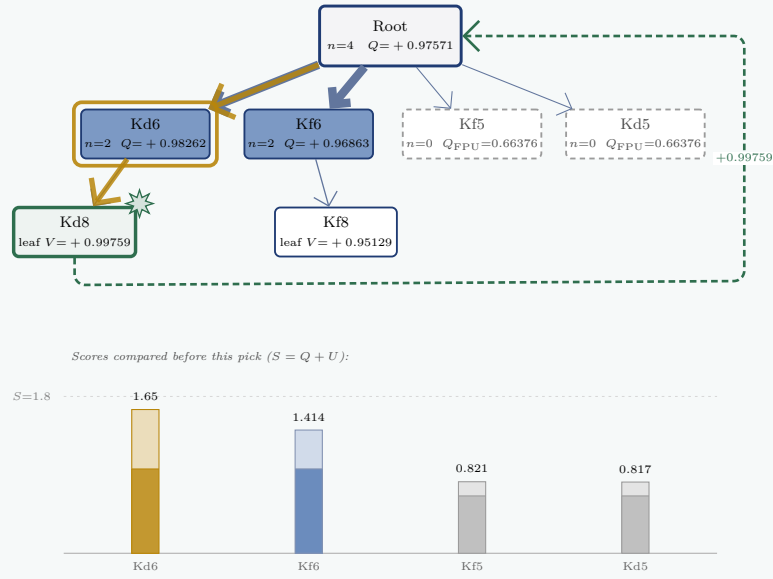


Figure 2.5: FIG-2.4: Iteration 4. Kd6 selected ($S = 1.64983$). Selection recurses down Kd6 branch to new leaf Kd8 (depth 2). Leaf returns +0.99759, updating Kd6's average Q to +0.98262.

Real engine output: Iteration 5

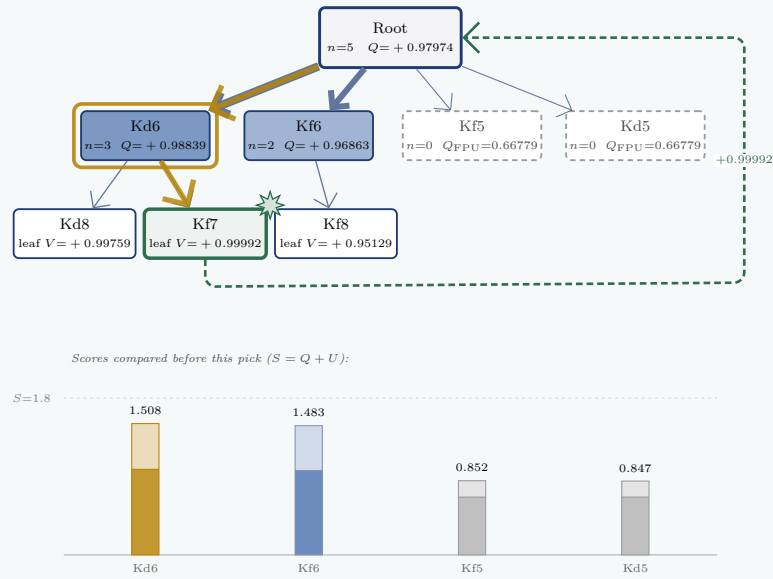


Figure 2.6: FIG-2.5: Iteration 5. Kd6 selected ($S = 1.50779$). Selection expands Kf7 under Kd6 (depth 2). Leaf returns +0.99992. Kd6 subtree widens.

Real engine output: Iteration 6

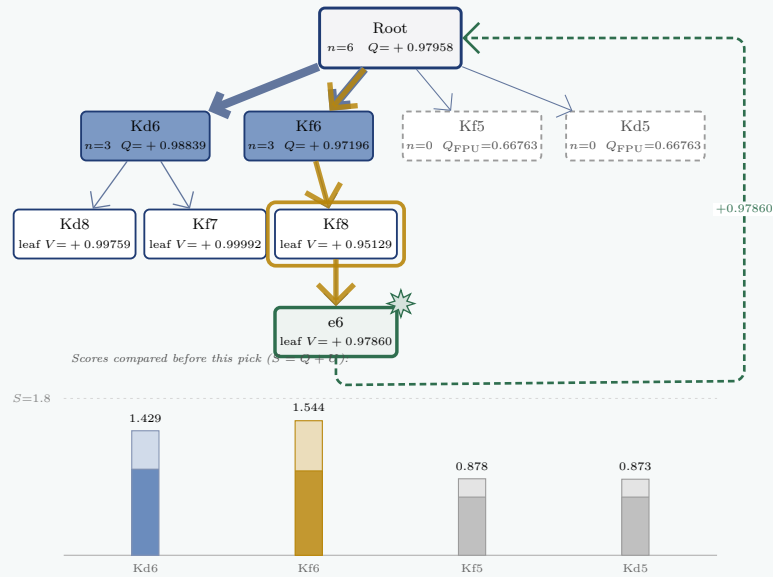


Figure 2.7: FIG-2.6: Iteration 6. Kf6 selected ($S = 1.54411$). Selection reaches depth 3 at e6 (under Kf8). Leaf returns +0.97860. Search deepens autonomously.

Real engine output: Iteration 7

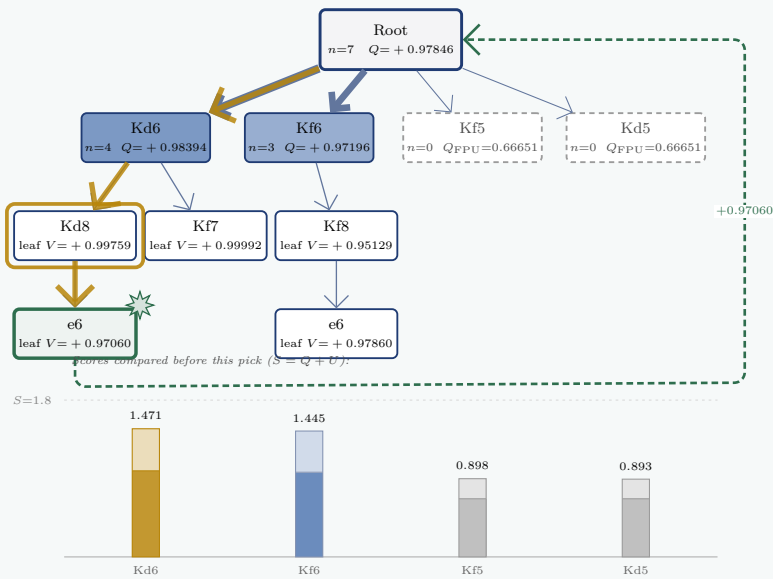


Figure 2.8: FIG-2.7: Iteration 7. Kd6 selected ($S = 1.47084$). Selection reaches depth 3 at e6 (under Kd8). Leaf returns +0.97060.

Real engine output: Iteration 8

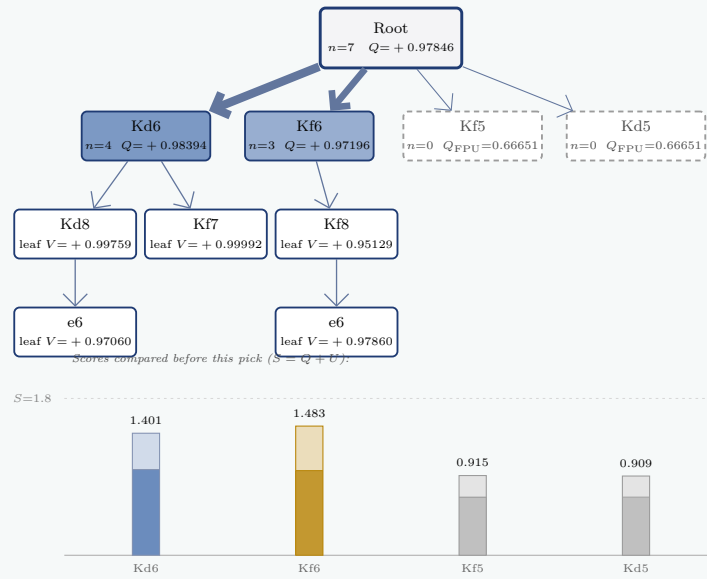


Figure 2.9: FIG-2.8: Iteration 8 (Verification). State after 7 backed-up values. Hand arithmetic matches lc0.exe go nodes 8 visit counts and Q values to 5 decimal places.

Real engine output: Search Time-Lapse

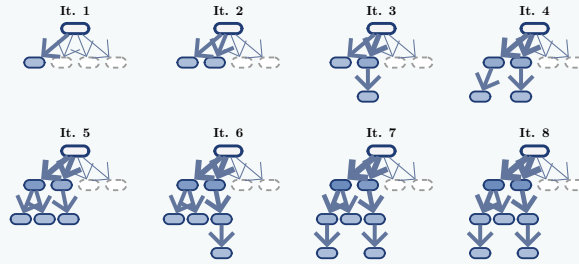
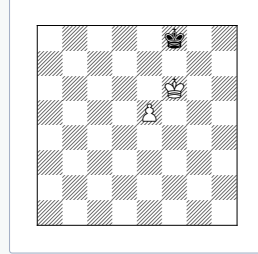


Figure 2.10: FIG-2.9: The whole search time-lapse. Sequence of tree states across iterations 1 to 8 showing structural growth.

2.2 The Sign Flip Convention

Evaluations are recorded from the point of view of the player to move at that node. As values walk back up the tree, the sign flips at every ply:

Real engine output: Sign Flips



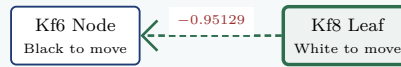
Leaf Node (after 1.Kf6 Kf8)

Side to move: **White**

Evaluated $V = +0.95129$

"White has a winning advantage (+0.951)."

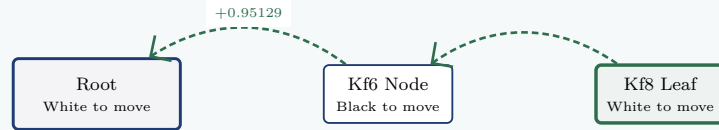
(a) Leaf Evaluation (White POV)



Incoming Edge to Kf8 (Black to move)

Black evaluates White's $+0.95129$ win as -0.95129 for Black.

(b) Black-to-move Edge



Root Edge (White to move)

Sign flips back: White receives $+0.95129$.

Kf6 running average: $(0.98598 + 0.95129)/2 = 0.968635$.

(c) Root Edge (White POV)

Figure 2.12: FIG-2.10: Iteration 3 Sign Flips. New leaf after 1.Kf6 Kf8 gives $+0.95129$ (White). Black edge receives -0.95129 . Root edge receives $+0.95129$, updating Kf6 average to 0.968635 .

2.3 Emergent Depth

Established mechanism: Autonomous Search Expansion

Depth Emergence across Iterations

Iterations 1–2: Stop at root's children (**depth 1**)

Iterations 3–5: Reach **depth 2** (Kf8, Kd8, Kf7 appear)

Iterations 6–7: Reach **depth 3** (e6 pawn moves appear)

Leela has no depth parameter at all. Depth is what happens when U collapses.

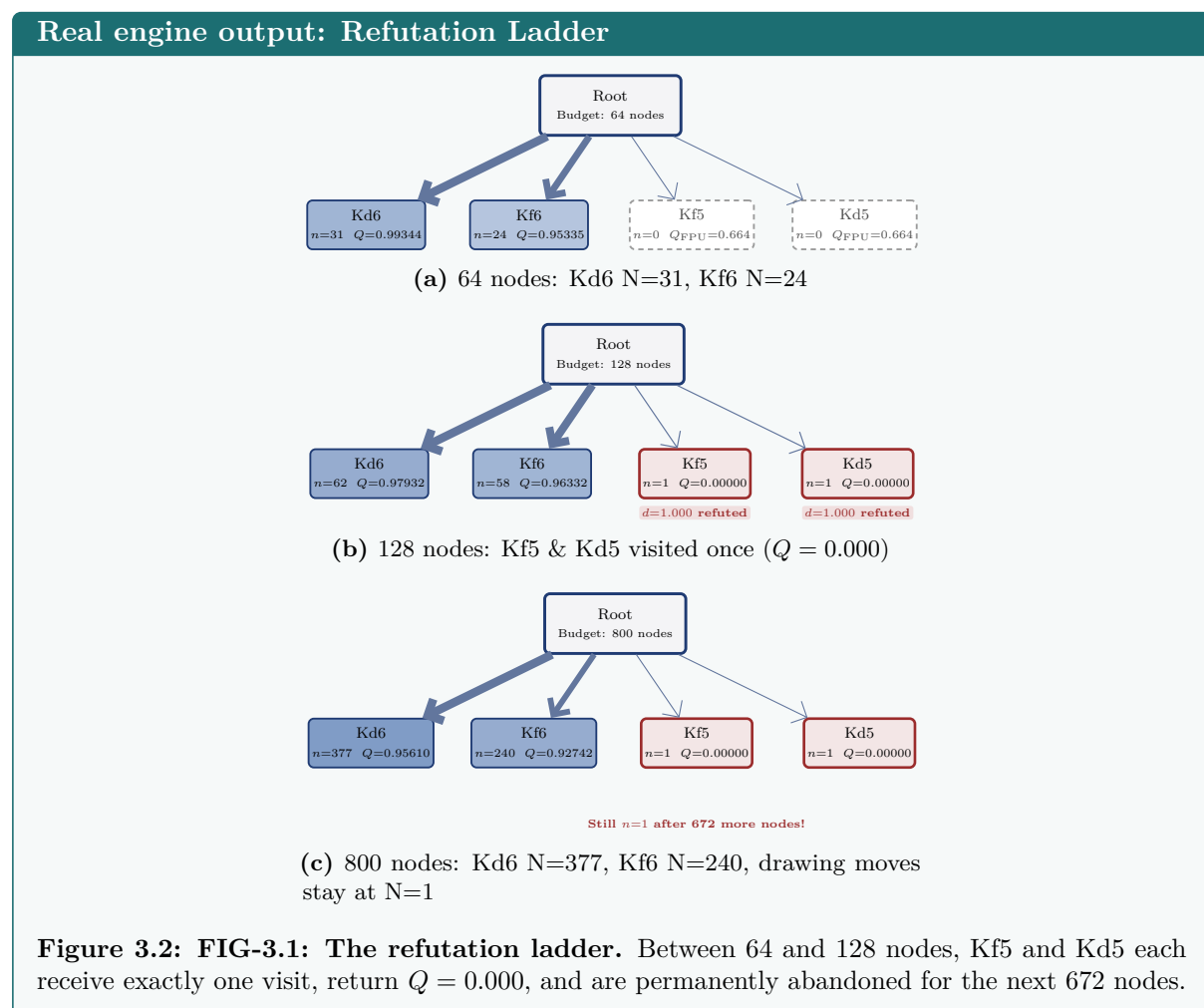
Figure 2.13: FIG-2.11: Depth Emergence. Iterations 1–2 stop at depth 1; 3–5 reach depth 2; 6–7 reach depth 3. Depth is an emergent outcome of U decay, not a pre-configured parameter.

Chapter 3

Why the Search Abandons a Move

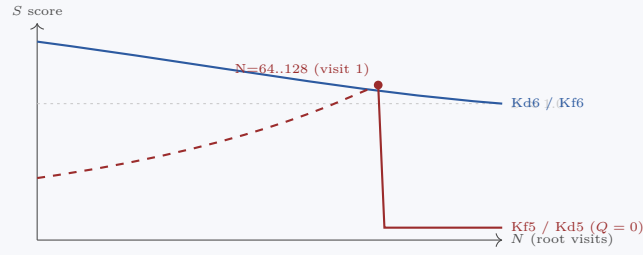
We examine how the search handles bad candidate moves across expanded visit budgets.

3.1 The Refutation Ladder



3.2 The Score Decay Race

Established mechanism: PUCT Score Curves

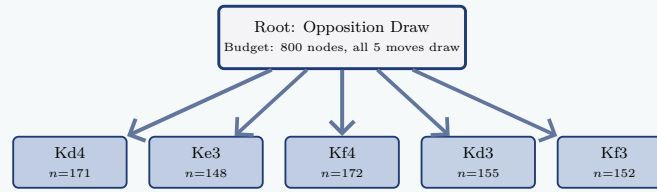


Curves derived via PUCT formula from measured priors; markers at measured $N = 64, 128$.

Figure 3.3: FIG-3.2: PUCT score race. Solid markers represent measured engine budgets ($N = 64, 128$). Dashed curves represent theoretical $S(N) = Q_{\text{FPU}}(N) + U(N)$ derived via the PUCT formula as unvisited U grows with $\sqrt{N}$ until a single visit drops Q to zero.

3.3 Contrast: When Nothing Matters

Real engine output: Dead Drawn Positions



Flat visit distribution across all legal moves: position is drawn.

Figure 3.4: FIG-3.3: Opposition Draw Contrast. In a dead drawn position, visit counts split almost evenly across all 5 legal moves ($N = 171, 148, 172, 155, 152$). A flat tree indicates no tactical differentiation.

3.4 The Honest Caveat

Established mechanism: Single-Look vs Deep Refutation

Refutation in 1 Look

Value head catches draw/loss immediately on first visit ($Q = 0.000$). Search abandons move.

Refutation 5 Plies Deep

Value head initial look is optimistic. Search must build subtree before truth emerges.

Figure 3.5: FIG-3.4: One-look vs deep refutation. Rapid refutation relies on the value head recognizing drawn/lost structures on the first visit. Deep tactical refutations require building multi-ply subtrees.

Chapter 4

Reading What the Engine Decided

We analyze how visit counts and evaluation proofs determine the final played move.

4.1 The Moment the Engine Changed Its Mind

In the Morphy position (4kb1r/p2n1ppp/4q3/4p1B1/4P3/1Q6/PPP2PPP/2KR4 w k - 0 16), white has a forced mate in 2 starting with 16.Qb8+!

Real engine output: Morphy Mind Change

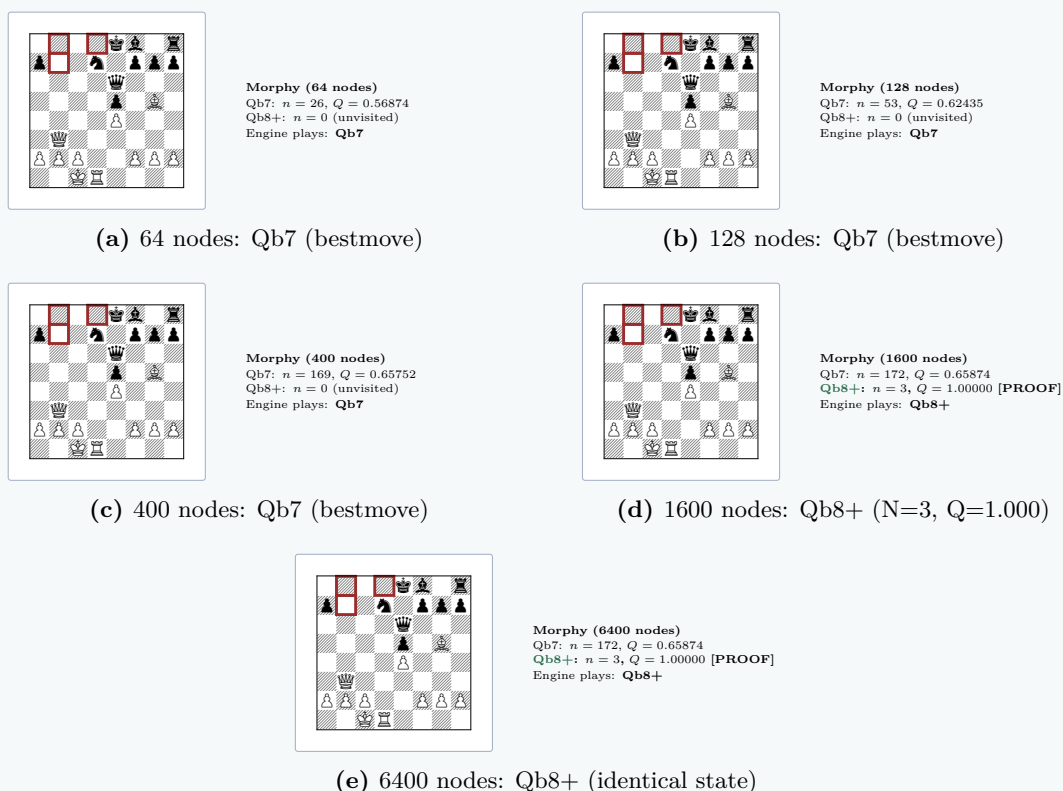


Figure 4.2: FIG-4.1: Mind change progression. Through 400 nodes, Qb8+ has zero visits. At 1600 nodes, 3 visits yield $Q = 1.000$ (proof of mate), causing bestmove to flip from Qb7 to Qb8+. At 6400 nodes, state remains identical.

4.2 Proof Beats Sampling

Real engine output: Exact Proofs vs Statistical Sampling



Proof beats sampling: A move visited 3 times with a proven mate outranks a move sampled 172 times with $Q = 0.659$.

Figure 4.3: FIG-4.2: Proof outranks sampling. A move visited 3 times with a proven mate ($Q = 1.000$) outranks a move sampled 172 times with $Q = 0.659$.

4.3 Policy Proposes, Value Disposes

Real engine output: Policy and Value Interaction

Root Policy Distribution (K+P position)

Winning moves: 89.36% (Kd6, Kf6) 10.64% Kf5, Kd5

Value head on the 10.64% slice: $Q = 0.000$ (1 visit refutation)

Figure 4.4: FIG-4.3: Policy proposes, value disposes. Policy assigns 89.36% to winning moves and 10.64% to draws. Value head scores the drawing slice 0.000 on first visit.

4.4 Which Number Chooses the Move?

Established mechanism: Visit Ranking vs Evaluation



Exception: Proven wins ($Q = 1.000$) override visit sampling.

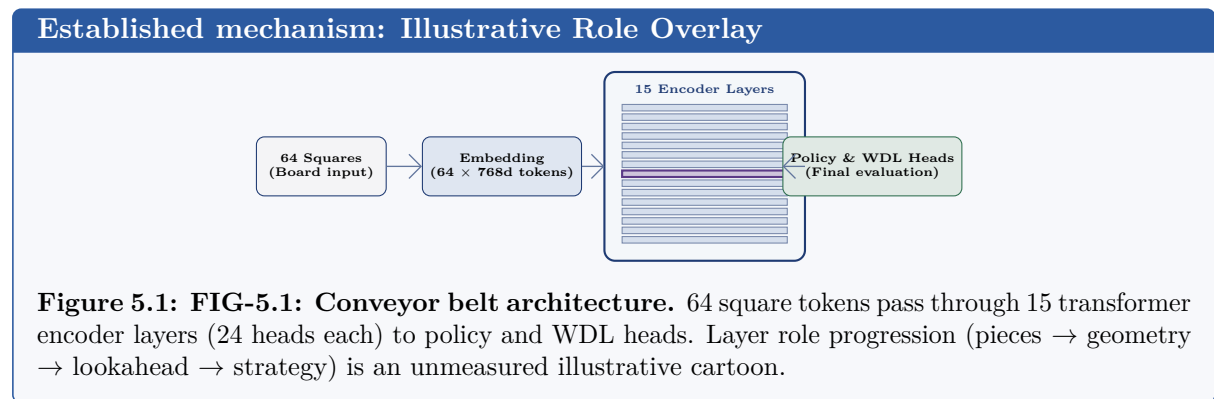
Figure 4.5: FIG-4.4: Move decision flowchart. Candidates are ranked by visit count n . High Q on low n is a question, not an answer, unless $Q = 1.000$ (proof).

Chapter 5

Inside the Network

This chapter details internal network processing, clearly separating established architecture and measured outputs from unverified hypotheses.

5.1 The Conveyor Belt



5.2 The Linear Probe Hypothesis

This project's hypothesis — not yet verified: Linear Probe Model

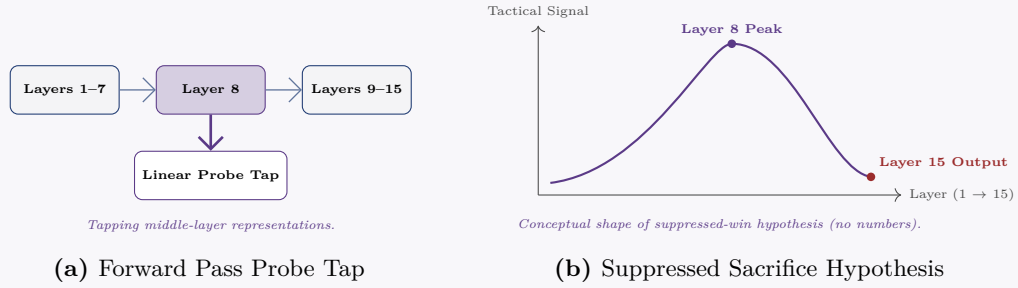


Figure 5.3: FIG-5.2: Probe hypothesis. Conceptual model of middle-layer tactical awareness suppressed by final policy output. No fabricated internal probe values are used.

5.3 Measured Counterpart: Policy Suppression

Real engine output: Policy Suppression of Forced Mate

Real Engine Output: Morphy Position Policy Pass

$P(Qb8+) = 1.60\%$ (*Network instinct barely considered the mate*)

$Q(Qb8+) = 1.00000$ (*Search found mate at $n = 3$ and overrode policy*)

Measured proof of policy override without needing an internal probe hypothesis.

Figure 5.4: FIG-5.3: Measured instinct override. In the Morphy position, $P(Qb8+) = 1.60\%$. The instinct nearly missed a forced mate in 2, providing a measured counterpart to policy suppression without probe assumptions.

5.4 Attention Map Visualization Tooling

This project's hypothesis — not yet verified: Tooling Limitation & Target Concept

Current Visual Tooling
Attention averaged over all 15 layers $\times$ 24 heads (diffuse map).

Single Head Target
Single head attention map (future goal — drawn hollow).

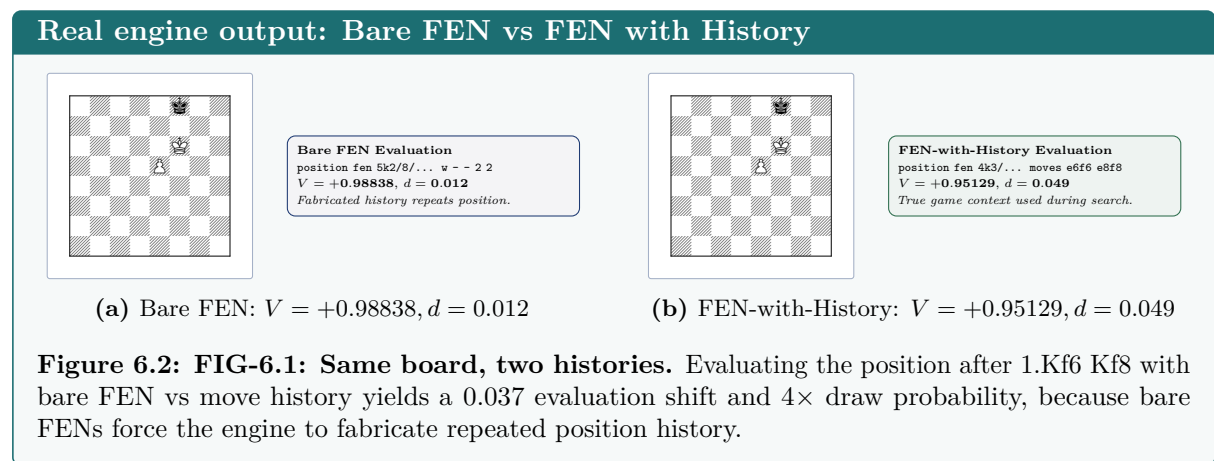
Figure 5.5: FIG-5.4: Attention maps. Current visualizations average attention across all 15 layers and 24 heads (diffuse map). Single-head isolation remains a target concept.

Chapter 6

Traps for System Developers

We highlight critical technical gotchas when building software tools on top of LC0 evaluations.

6.1 The FEN-vs-History Discrepancy



6.2 Closing Summary Card

- Product Visual Guide Summary**
- **Policy Arrows** → Prior attention (FIG-1.3)
 - **Variations** → Visit-weighted tree paths (FIG-2.9)
 - **Refuted Moves** → One-visit $Q = 0$ drops (FIG-3.1)
 - **High Q, Low n** → Questions, not answers (FIG-4.4)

Figure 6.3: FIG-6.2: Product visual guide mapping summary card. Reference mapping UI display features to underlying search engine mechanics.